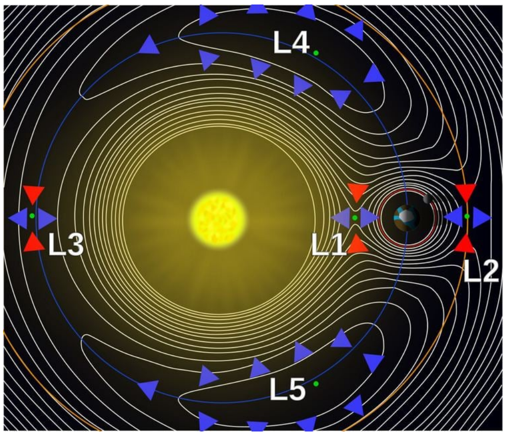

Figure 1: Hill Spheres in the CR3BP

Temporary Capture in the Earth-Moon Circular Restricted Three-Body Problem (CR3BP)

ASEN 6062: Celestial Mechanics

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1 Abstract

This paper studies the temporary ballistic capture of Moon satellites in the Earth-Moon CR3BP system. I will derive the equations of motion from Lagrangian mechanics and reformulate them using Hamiltonian mechanics to analyze the system's structure and conserved quantities, including the Jacobi Integral. Numerical simulations are used to identify trajectories that exhibit temporary capture behavior, which will be interpreted through zero-velocity curves and phase space geometry. The focus will be on understanding how invariant structures and energy constraints govern capture and escape in a non-trivial three-body system.

2 Nomenclature

H	Hamiltonian
L	Lagrangian
n	Mean motion
r	Orbital radius
T	Kinetic energy
t_0, t_f	Initial and final time
U	Potential energy
V	Orbital velocity magnitude
$\mathbf{x}$	State vector, $[\vec{x}, \dot{\vec{x}}, \mu]^T$
θ	Angular position
μ	Normalized mass of the primary body (m2/(m1=m2)
ω	Angular velocity

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The solution to finding the trajectories of particles which result in a temporary lunar capture can be best modeled using the Circular Restricted Three Body Problem (CR3BP)[**Brouke**] in a uniformly rotating (synodic) frame. The CR3BP models the planar motion of a massless test particle under the gravitational influence of two primary bodies with masses m_1 and m_2 where $m_1 > m_2$. The primaries are assumed to move on circular orbits about their common barycenter, while the third body has negligible mass and therefore does not perturb the motion of the primaries. This idealization captures key dynamics of systems with a strong mass hierarchy (e.g., spacecraft motion in the Earth–Moon or Sun–Earth environments), while retaining the essential non-integrability of three-body gravitational motion; in general, trajectories of the test particle must be propagated numerically. Analysis is performed in a synodic reference frame whose origin is fixed at the system barycenter and whose angular velocity matches the mean motion of the primaries so that both primaries appear stationary. The distance between the primaries is set to unity, the sum of the primary masses is unity and the system is characterized by the mass parameter $\mu = \frac{m_2}{m_1+m_2}$. In this frame, the primaries lie on the x-axis at fixed locations (-μ, 0, 0) and (1-μ, 0, 0) while a point co-rotating with the frame has zero velocity. The resulting equations of motion include both Coriolis and centrifugal terms.

5 Massless Particle Dynamics

5.1 Motion in Inertial Frame

[**Brouke**] In an inertial frame defined with the barycenter of the primary bodies as the origin, the location of the infinitesimal third body is defined by:

$$r_1^2 = (x_1 - x)^2 + (y1 - y)^2 + z^2 = (x + \mu)^2 + y^2 + z^2 \tag{1}$$

$$r_2^2 = (x_2 - x)^2 + (y_2 - y)^2 + z^2 = (x - 1 + \mu)^2 + y^2 + z^2 \tag{2}$$

where x and y are the coordinates of the third body, x_1 and y_1 are the coordinates of the primary, x_2 and y_2 are the coordinates of the secondary, r_1 is the relative distance between the first and third bodies and r_2 is the relative distance between the second and third bodies and μ is the mass parameter. The equations of motion for the third body in this inertial frame are as follows:

$$\ddot{x} - 2\dot{y} = x - \frac{1-\mu}{r_1^3}(x + \mu) - \frac{\mu}{r_2^3}(x - 1 + \mu) \tag{3}$$

$$\ddot{y} + 2\dot{x} = y - \frac{1-\mu}{r_1^3}y + \frac{\mu}{r_2^3}y \tag{4}$$

$$\ddot{z} = -\frac{1-\mu}{r_1^3}z - \frac{\mu}{r_2^3}z \tag{5}$$

5.2 Lagrangian Derivation

In an inertial frame, the gravitational potential depends explicitly on time because the primaries are moving. Consequently, the Lagrangian is time-dependent and the system does not have a conserved energy integral nor time-invariant equilibrium points in inertial coordinates. The Lagrangian in the inertial frame, L_I, is:

$$L_I = f(\boldsymbol{r}, \dot{\boldsymbol{r}}, t) = T - U = \frac{1}{2}||\dot{r}^2|| + \frac{\mu_1}{||\boldsymbol{r} - \boldsymbol{r_1}(t)||} + \frac{\mu_2}{||\boldsymbol{r} - \boldsymbol{r_2}(t)||} \tag{6}$$

where T is the kinetic energy, U is the potential energy, $\boldsymbol{r}$ is the position of the test particle and $\boldsymbol{r_1}$ and $\boldsymbol{r_2}$ are the positions of the primaries varying over time. To obtain an autonomous formulation, the system is transformed into a synodic frame as defined in the previous section. In this frame, the angular velocity $\boldsymbol{\Omega} = n\hat{z}$ where n is the mean motion of the primaries and is equal to one. The system is parameterized by μ, primaries are fixed and the state of the test particle must be numerically computed over time. In this frame,

$$\boldsymbol{r_1} = (-\mu, 0, 0) \tag{7}$$
$$\boldsymbol{r_2} = (1-\mu, 0, 0) \tag{8}$$
$$\boldsymbol{q} = (x, y, z) \tag{9}$$
$$\dot{\boldsymbol{q}} = \dot{\boldsymbol{r_I}} + \boldsymbol{\Omega} \times \boldsymbol{r} \tag{10}$$

where $\boldsymbol{r_1}$ and $\boldsymbol{r_2}$ are fixed at their locations, $\boldsymbol{q}$ and $\dot{\boldsymbol{q}}$ are the position and velocity of the third body, respectively, and $\dot{\boldsymbol{r_I}}$ is the velocity of the third body in the inertial frame. With $||\boldsymbol{\Omega}|| = 1$ and $\dot{\Omega} = 0$ the Lagrangian then becomes time invariant:

$$L = f(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \frac{1}{2}||\dot{\boldsymbol{q}}||^2 + \boldsymbol{\Omega} \times (\boldsymbol{q} \times \dot{\boldsymbol{q}}) + \frac{1}{2}||\boldsymbol{\Omega} \times \boldsymbol{q}||^2 + \frac{1-\mu}{r_1} + \frac{\mu}{r_2} \tag{11}$$
$$L = \frac{1}{2}(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) + (x\dot{y} + \dot{x}y) + \frac{1}{2}(x^2 + y^2) + \frac{1-\mu}{r_1} + \frac{\mu}{r_2} \tag{12}$$

where $\boldsymbol{q}$, $\dot{\boldsymbol{q}}$, r_1, and r_2 are defined as above. The equations of motion are then derived from the time derivative of the Lagrangian:

$$\frac{d}{dt}(\frac{\partial L}{\partial \dot{\boldsymbol{q}}}) - \frac{\partial L}{\partial \boldsymbol{q}} = 0 \tag{13}$$

and become:

$$\ddot{x} - 2\dot{y} = x - \frac{1-\mu}{r_1^3}(x+\mu) - \frac{\mu}{r_2^3}(x - 1 + \mu) \tag{14}$$
$$\ddot{y} + 2\dot{x} = y - \frac{1-\mu}{r_1^3}y - \frac{\mu}{r_2^3}y \tag{15}$$
$$\ddot{z} = -\frac{1-\mu}{r_1^3}z - \frac{\mu}{r_2^3}z \tag{16}$$

where (x, y, z) are the coordinates, $(\dot{x}, \dot{y}, \dot{z})$ and $(\ddot{x}, \ddot{y}, \ddot{z})$ are the respective velocity and acceleration of the third body at any point in time.

5.3 Hamilton Derivation

The Hamiltonian form defines the dynamical system's solution properties and reveals the structure of the CR3BP. It is central to analyzing conserved quantities and phase-space dynamics. The Hamiltonian is obtained via a Legendre transformation of the synodic-frame Lagrangian, equation 13. The canonical momenta, p, are associated with the coordinates, $\boldsymbol{q}$, defined as $\boldsymbol{p} = \frac{\partial L}{\partial \dot{\boldsymbol{q}}}$. The momenta are:

$$p_x = \dot{x} - y \tag{17}$$
$$p_y = \dot{y} + x \tag{18}$$
$$p_z = \dot{z} \tag{19}$$

with velocities expressed as:

$$\dot{x} = p_x + y, \dot{y} = p_y - x, \dot{z} = p_z \tag{20}$$

The Hamiltonian is defined by:

$$H(\boldsymbol{q}, \boldsymbol{p}) = \boldsymbol{p} \cdot \dot{\boldsymbol{q}} - L \tag{21}$$

Substituting the above relations yields:

$$H = \frac{1}{2}(p_x^2 + p_y^2 + p_z^2) + p_x y - p_y x - \frac{1-\mu}{r_1} - \frac{\mu}{r_2} \tag{22}$$

where r_1 and r_2 are defined in equations (1) and (2). The Hamiltonian takes on the form:

$$\dot{\boldsymbol{q}} = \frac{\partial H}{\partial \boldsymbol{p}} = \boldsymbol{p} - \boldsymbol{\Omega} \times \boldsymbol{q} \tag{23}$$

$$\dot{\boldsymbol{p}} = -\frac{\partial H}{\partial \boldsymbol{q}} = -\boldsymbol{\Omega} \times \boldsymbol{p} + \frac{\partial U}{\partial \boldsymbol{q}} \tag{24}$$

Explicitly,

$$\dot{p}_x = \frac{\partial H}{\partial x} = p_y - [(1-\mu)\frac{x+\mu}{r_1^3} + \mu\frac{x-1+\mu}{r_2^3}] \tag{25}$$

$$\dot{p}_y = -\frac{\partial H}{\partial y} = -p_x - [(1-\mu)\frac{y}{r_1^3} + \mu\frac{y}{r_2^3}] \tag{26}$$

$$\dot{p}_z = \frac{\partial H}{\partial z} = -[(1-\mu)\frac{z}{r_1^3} + \mu\frac{x-1+\mu}{r_2^3}] \tag{27}$$

and $\dot{\boldsymbol{q}}$ is defined in equation XX. Differentiating the position equations and substituting the momenta equations restores the equations of motion defined in Section 5.2, confirming consistency between the Lagrangian and Hamiltonian formulations.

5.4 Jacobi Integral and Zero Velocity Curves

Because the Lagrangian is time-invariant in the synodic frame, equation (12), the Legendre transform yields a conserved integral of motion, called the Jacobi integral. The Hamiltonian, H, is constant along trajectories where $\frac{dH}{dt} = 0$. To relate this to the Jacobi integral, the effective potential is defined as:

$$U(x, y, z) = \frac{1}{2}(x^2 + y^2) + \frac{1-\mu}{r_1} + \frac{\mu}{r_2} \tag{28}$$

Expressing the Hamiltonian in terms of velocity yields:

$$H = \frac{1}{2}v^2 - U(x, y, z) \tag{29}$$

Multiplying that by -2 defines the Jacobi integral:

$$C = 2U(x, y, z) - v^2 \tag{30}$$

which relates to the Hamiltonian by:

$$C = -2H \tag{31}$$

Motion is restricted to regions where $2U(x, y, z) \geq C$. The boundary $2U(x, y, z) = C$ defines the zero-velocity surface (ZVS) which yields the zero-velocity curves (ZVCs) when z=0. Shown in figures 2 and 3. These curves define accessible and forbidden regions for given values of the Jacobi constant. These regions constrain transport trajectories between the Earth and Moon and govern capture and escape through the L1 and L2 gateways.

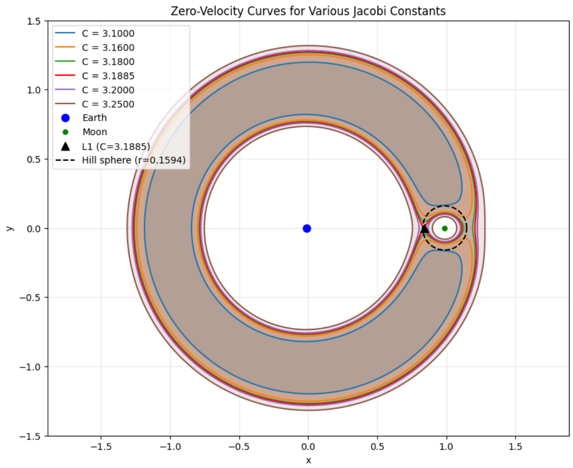

Figure 2: Zero Velocity Curves for Various Jacobi Constants in the Earth-Moon System

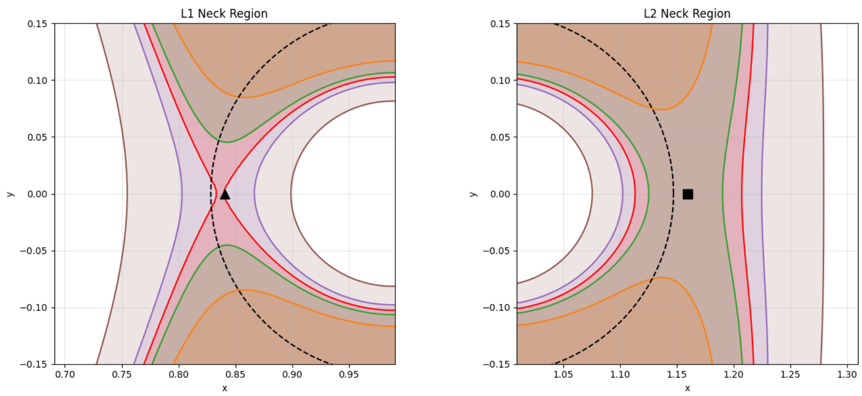

Figure 3: ZVCs near L1 and L2 Necks

Now that the global energy constraints are known, a definition of temporary lunar capture is established. In planetary dynamics, a particle is said to be temporarily captured by the secondary body if it enters the Moon's Hill sphere with sufficiently low relative velocity such that it becomes gravitationally bound for a finite duration before eventual escape. The energy is given by:

$$E_{rel} = \frac{1}{2}v_{rel}^2 - \frac{(1-\mu)}{r_{rel}} \tag{32}$$

A trajectory is classified as temporarily captured when $E_{rel} < 0$ over a continuous time interval while the particle remains within the lunar Hill sphere.

6 State Transition Matrix

While the Jacobi integral and ZVCs provide global constraints on the massless third body, they do not characterize the local behavior of a trajectory at any particular point. ZVCs identity if motion is permitted or forbidden in a region, but they do not give any information on the actual trajectory or how neighboring trajectories evolve. Temporary lunar capture can occur in regions where the ZVCs allow access to the Moon's Hill sphere through the L1 and L2 gateways but do not guarantee long-term capture. This behavior is governed by the local stability properties of the system solutions. To quantify this local behavior, the state transition matrix (STM) of the linearized system along a nominal trajectory is evaluated. The STM identifies the stability of the system to perturbations and helps distinguish temporary capture from escape dynamics.

6.1 STM Derivation

At this point, the system will be restricted to the planar CR3BP to simplify the analysis of the STM which removes the z coordinates. Hamilton's equations:

$$\boldsymbol{x} = \begin{bmatrix} \boldsymbol{q} \\ \boldsymbol{p} \end{bmatrix}, \quad \boldsymbol{q} = (x, y), \quad \boldsymbol{p} = (p_x, p_y) \tag{33}$$

can be written as:

$$\dot{\boldsymbol{x}} = J\nabla H(\boldsymbol{x}) \tag{34}$$

where J is the symplectic matrix:

$$\begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix} \quad I = I_{2x2} \tag{35}$$

Linearizing the equations of motion about a nominal solution with small perturbations, $\delta\boldsymbol{x}(t)$ gives:

$$\delta\dot{\boldsymbol{x}} = \frac{\delta}{\delta x}(J\nabla H(x))\Big|_{x(t)}\delta x \tag{36}$$

$$= J\nabla^2 H(x(t))\delta x \tag{37}$$

where $\nabla^2 H$ is the Hessian of the Hamiltonian:

$$\nabla^2 H = \begin{bmatrix} H_{qq} & H_{qp} \\ H_{pq} & H_{pp} \end{bmatrix} = \begin{bmatrix} H_{xx} & H_{xy} & 0 & -1 \\ H_{xy} & H_{yy} & 1 & 0 \\ 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix} \tag{38}$$

where $H_{pp} = I$, H_{qp} is the Coriolis coupling and H_{qq} contains the second derivative of the effective potential U_{xx}, U_{yy} and U_{xy}. The STM $\Phi(t)$ maps initial perturbations:

$$\delta\boldsymbol{x}(t) = \Phi(t, t_0)\delta\boldsymbol{x}(t_0)$$

(39)

and satisfies:

$$\dot{\Phi}(t) = A(t)\Phi(t), \quad \Phi(t_0) = I_{2x2} \tag{40}$$

where the Jacobian is:

$$A(t) = J\nabla^2 H(\boldsymbol{x}(t)) \tag{41}$$

with the structure:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ -H_{xx} & -H_{xy} & 0 & 1 \\ -H_{yx} & H_{yy} & -1 & 0 \end{bmatrix} \tag{42}$$

(43)

where the gradients of the effective potential are:

$$H_{xx} = (\frac{1}{r_1^3} - 3\frac{(x+\mu)^2}{r_1^5}) + \mu\Big(\frac{1}{r_2^3} - 3\frac{(x-1+\mu)^2}{r_2^5}\Big) \tag{44}$$

$$H_{yy} = (1-\mu)\Big(\frac{1}{f_1^5}\frac{1-\mu}{r_1^3}\left(1 - 3\frac{(x+\mu)^2}{+}r_1^2\right) - \frac{\mu}{r_2^3}\left(1 - 3\frac{(x-1+\mu)^2}{r_2^2}\right) \tag{45}$$

$$U_{yy} = 1 - \frac{1-\mu}{r_1^3}\left(1 - 3\frac{y^2}{r_1^2}\right) - \frac{\mu}{r_2^3}\left(1 - 3\frac{y^2}{r_2^2}\right) \tag{46}$$

$$U_{xy} = U_{yx} = 3(1-\mu)\frac{(x+\mu)y}{r_1^5} + 3\mu\frac{(x-1+\mu)y}{r_2^5} \tag{47}$$

Using this method, it can be proven that A is a symplectic matrix with:

$$A = J\nabla^2 H, \quad (\nabla^2 H)^T = \nabla^2 H \tag{48}$$

$$\Phi(t)^T J \Phi(t) = J \tag{49}$$

6.2 The L_1 and L_2 Equilibrium Points

Based on research by Paez and Guzzo, [**Paez**], slow close encounters of a third body captured by the secondary mass can be studied in the framework of the dynamics at the Lagrangian points L_1, L_2 of the CR3BP. This system is evaluated in the neighborhoods of the these equilibrium points while the eigenvalues of the STM are evaluated to understand the stability of the local system. The Hamiltonian structure guarantees that the eigenvalues come in $(\lambda, -\lambda, \overline{\lambda}, -\overline{\lambda})$ pairs. At L_1 and L_2 there are one real pair, $\pm\alpha$, and one imaginary pair $\pm i\beta$. By definition the equilibrium point in the rotating frame is:

$$U_x(x_L, 0) = 0, \quad U_y(x_L, 0) = 0 \tag{50}$$

with small perturbations:

$$x = x_L + \delta x, \quad y = 0 + \delta y, \quad |\delta x|, |\delta y| \ll 1 \tag{51}$$

From here, the first-order Taylor expansions result in:

$$U_x(x_L + \delta x) \approx U_{xx}(x_L, 0)\delta x + U_{xy}(x_L, 0)\delta y \tag{52}$$

$$U_y(x_L + \delta x) \approx U_{yx}(x_L, 0)\delta x + U_{yy}(x_L, 0)\delta y \tag{53}$$

7 Verification and numerical tests

Jacobi integral test (if applicable) STM symplectic or eigenvalue structure test [Project — PDF]

8 Solutions of interest + stability results

trajectories + parameter sweeps STM-based stability analysis [Project — PDF]

9 Discussion and future work

** Add discussion here on how the intial conditions were chosen. mention zvc and explain what allows entry into the moon Hill sphere and temporary capture, restricted to e-m system vs. escape.

Captured definition... In planetary dynamics, a particle is considered temporarily captured if it: 1) enters a body's Hill sphere with sufficiently low relative velocity, such that 2) it becomes gravitationally bound for 3) a finite period of time. If the two-body energy relative to the secondary body is:

$$E_{rel} = \frac{1}{2}v_{rel}^2 - \frac{(1-\mu)}{r_{rel}} \tag{54}$$

Capture Metrics Minimum recommended:

Particle inside Moon's Hill sphere Moon-relative Kepler energy ¡ 0 Residence time T¿TminT ¿ $\mathrm{T}_{min}(e.g., multiplemoonrevolutions)$

10 Stability of XXX in the Earth-Moon CR3BP System

11 Simulation and Results

12 Conclusion